Zhiqi Gao

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Bio

I am a fourth-year Ph.D. student in Computer Science at the University of Wisconsin-Madison, advised by Professor Frederic Sala. My research takes a data-centric approach to large language models (LLMs) and LLM agents, focusing on (i) evaluating them faithfully through verifiable outcomes, understanding skills, and metric design, and (ii) improving them through data selection, curation, and data synthesis.

Education

University of Wisconsin-Madison **Sep. 2023 – Expected Aug. 2028**
Ph.D. in Computer Science *Madison, Wisconsin*

University of Wisconsin-Madison **Sep. 2023 – Aug. 2026**
M.S. in Computer Science *Madison, Wisconsin*

University of Wisconsin-Madison **Sep. 2019 – May 2023**
B.S. in Computer Science & Mathematics *Madison, Wisconsin*

Publications

Conferences & Journals

Understanding and Dating LLM-Era Stylistic Changes via the Cohort Deviation Measure
Zhiqi Gao*, Frederic Sala*

- Under review at NAACL 2027

Models Can Model, But Can't Bind: Structured Grounding in Text-to-Optimization
Zhiqi Gao* , *Albert Ge** , *Alexander Michael Berenbeim, Nathaniel D. Bastian, Frederic Sala*

- Conference on Language Modeling (COLM), 2026
- ICLR 2026 DATA-FM Workshop (Preliminary Version, as "OR-LLM-Bench")

Test-Time Scaling Makes Overtraining Compute-Optimal
Nicholas Roberts, Sungjun Cho, ***Zhiqi Gao****, Tzu-Heng Huang, Albert Wu,* et al.*, Aws Albarghouthi, Frederic Sala*

- Conference on Language Modeling (COLM), 2026

Fine-Tuning Small Reasoning Models for Quantum Field Theory
Nathaniel S. Woodward, ***Zhiqi Gao****, Yurii Kvasiuk, Kendrick M. Smith, Frederic Sala, Moritz Münchmeyer*

- Machine Learning: Science and Technology (MLST), 2026 (Impact Factor 4.6)
- Conference on Physics and AI (PAI), 2026

Pretrained Hybrids with MAD Skills
Nicholas Roberts, Samuel Guo, ***Zhiqi Gao****, Satya Sai Srinath Namburi GNVV, Sonia Cromp,* et al.*, Frederic Sala*

- Conference on Language Modeling (COLM), 2025

Theoretical Physics Benchmark (TPBench)
– a Dataset and Study of AI Reasoning Capabilities in Theoretical Physics
Daniel J.H. Chung, ***Zhiqi Gao****, Yurii Kvasiuk, Tianyi Li, Moritz Münchmeyer, Maja Rudolph, Frederic Sala,* et al.

- Machine Learning: Science and Technology (MLST), 2025 (Impact Factor 4.6)
- Midwest Machine Learning Symposium (MMLS), 2025 (**Lightning Talk**)

Workshops

Test-time Scaling Techniques in Theoretical Physics – A Comparison of Methods on the TPBench Dataset
***Zhiqi Gao**[*], Tianyi Li[*], Yurii Kvasiuk,* et al.*, Frederic Sala, Moritz Münchmeyer*

- NeurIPS 2025 Machine Learning and the Physical Sciences (ML4PS) Workshop

Re-Structuring CLIP's Language Capabilities
***Zhiqi Gao**, Frederic Sala*

- Midwest Machine Learning Symposium (MMLS), 2025

Industry Experience

Roblox Corporation **May 2023 – Aug. 2023**
Software Engineer Intern *AI/ML Team*

- Designed, developed, and deployed a full-stack Slack bot that combines a vector database with large language models (LLMs) to answer complex questions over internal knowledge via retrieval-augmented generation (RAG), outperforming the company's existing Q&A Slack bot.
- Created an efficient data pipeline, ingesting diverse documents (Confluence, Stack Overflow, GitHub) and generating vector embeddings for rapid retrieval.

Technical Skills

Programming Languages: Python, Java, C/C++, SQL, Wolfram
ML Frameworks: PyTorch, Hugging Face Transformers, vLLM, DeepSpeed, Accelerate

Teaching Experience

University of Wisconsin-Madison **Aug. 2023 – Present**
Teaching Assistant *Madison, Wisconsin*

- CS 540, Introduction to Artificial Intelligence: mentor students on core AI concepts and machine learning basics (Fall 2024, Spring 2025, Spring 2026).
- CS 300, Programming II: help students understand object-oriented programming principles, advanced data structures, and debugging in Java (Fall 2023, Spring 2024, Fall 2026).

Service

Reviewer: NeurIPS 2024, 2025, 2026, COLM 2026, TMLR 2026